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Course: Compiler Design

Assignment Lab Task-4

Task: You are given an input file named **input.txt**. Each line contains **one statement**.

Your program must do the following:

Step A, Read and classify every line

For each line, classify it as one of these:

1. Variable Declaration
2. Function Declaration
3. Invalid

Write the classification for each line into **report.txt** in this format:

Line <n>: <classification>

Step B, Write only valid lines into a new file

Create **valid_code.txt** that contains only the lines that are classified as Variable Declaration or Function Declaration, in the same order.

Step C, Add a summary at the end of report.txt

At the end of report.txt, append:

- Total lines read
- Number of variable declarations
- Number of function declarations
- Number of invalid lines

Spaces may appear anywhere, but you can ignore spaces while checking.

1) Variable Declaration (only these forms)

Valid if it matches one of these patterns:

- int id;
- float id;
- double id;
- char id;

Where id must follow:

- first character is a letter or underscore
- remaining characters can be letters, digits, underscore

Examples:

- Valid: int x;, float _avg;, double a1;
- Invalid: int 2x;, int x y;, int x

2) Function Declaration (prototype, no parameters to keep it easy)

Valid if it matches:

- int fname();
- float fname();
- double fname();
- char fname();
- void fname();

Where fname follows the same identifier rule.

Examples:

- Valid: int sum();, void show();

• Invalid: int sum(int a);, int 3f();, int sum()

Sample input.txt

int x;
float avg;
int sum();
void show();
int 2x;
int f(int a);

Expected report.txt

Line 1: Variable Declaration
Line 2: Variable Declaration
Line 3: Function Declaration
Line 4: Function Declaration
Line 5: Invalid
Line 6: Invalid
Summary:
Total lines = 6
Variable Declarations = 2
Function Declarations = 2
Invalid = 2

Expected valid_code.txt

int x;
float avg;
int sum();
void show();

Answer:

```
#include <iostream>

#include <fstream>

#include <string>

#include <cctype>


using namespace std;


bool isIdentifier(string s)

{

   if (s.empty()) return false;
```

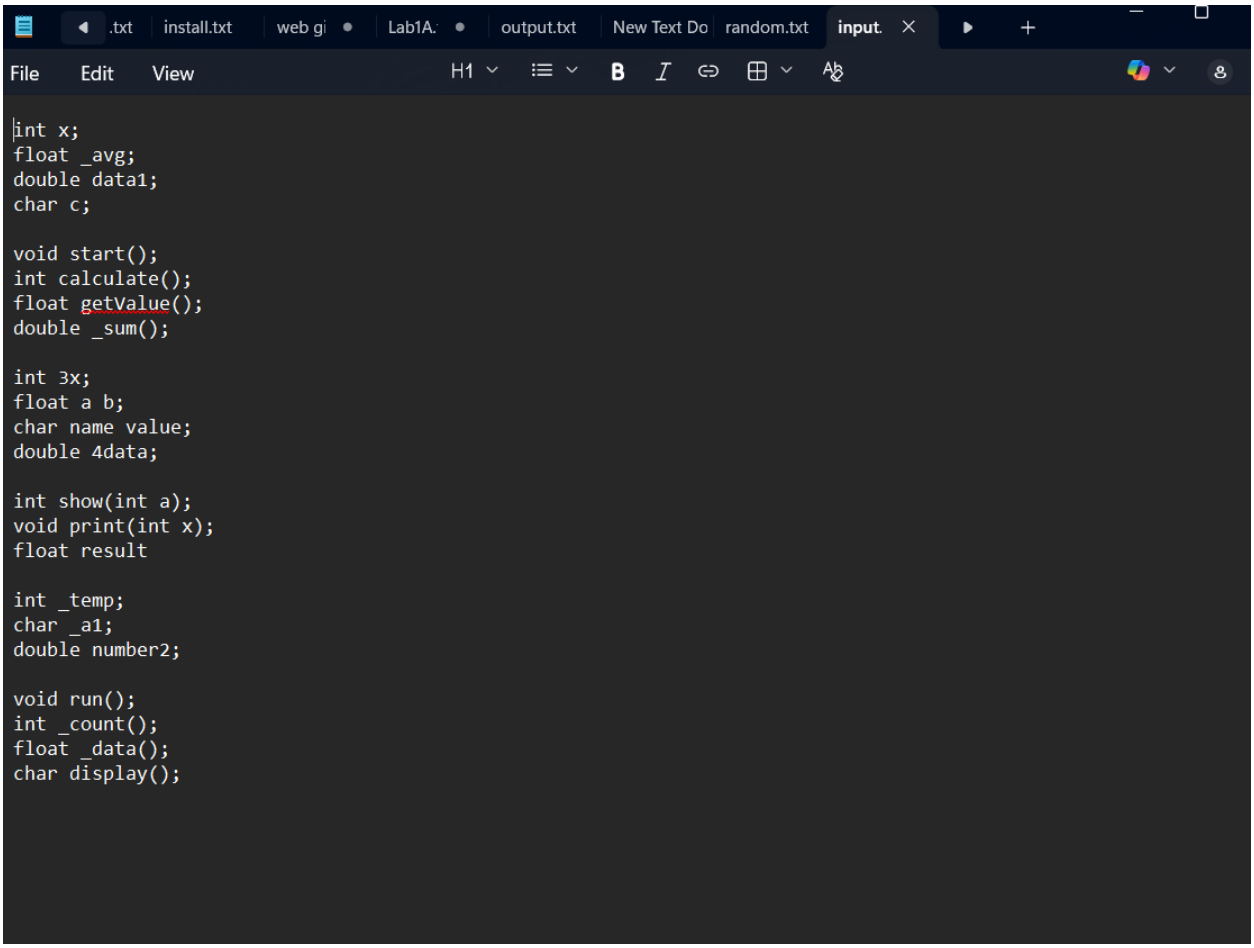
```
    if (!(isalpha(s[0]) || s[0] == '_')) return false;
    for (int i = 1; i < s.length(); i++)
    {
    if (!(isalnum(s[i])||s[i] == '_')) return false;
    }
    return true;
}
bool isVariable(string line)
{
    string type, id;
    int pos = line.find(' ');
    if (pos == -1) return false;
    type = line.substr(0, pos);
    id = line.substr(pos + 1);

    if (id.back() != ';') return false;
    id.pop_back();

    if (!(type == "int" || type == "float" || type == "double" || type == "char"))
        return false;

    return isIdentifier(id);
}
bool isFunction(string line)
{
    string type;
    int pos = line.find(' ');
    if (pos == -1) return false;
    type = line.substr(0, pos);
```

```
    string rest = line.substr(pos + 1);
if (!(type == "int" || type == "float" || type == "double" || type == "char" || type == "void"))
        return false;

int p1 = rest.find('(');
int p2 = rest.find(')');

    if (p1 == -1 ||p2 ==-1) return false;

    string fname = rest.substr(0, p1);

    if (rest.substr(p1) != "();") return false;

    return isIdentifier(fname);
}
int main()
{
    ifstream input("input.txt");
    ofstream report("report.txt");
    ofstream valid("valid_code.txt");

    if (!input.is_open())
    {
        cout << "input.txt open korte problem!" << endl;
        return 1;
    }
    string line;
    int lineNo = 0;
    int varCount = 0, funCount = 0, invalid = 0;
```

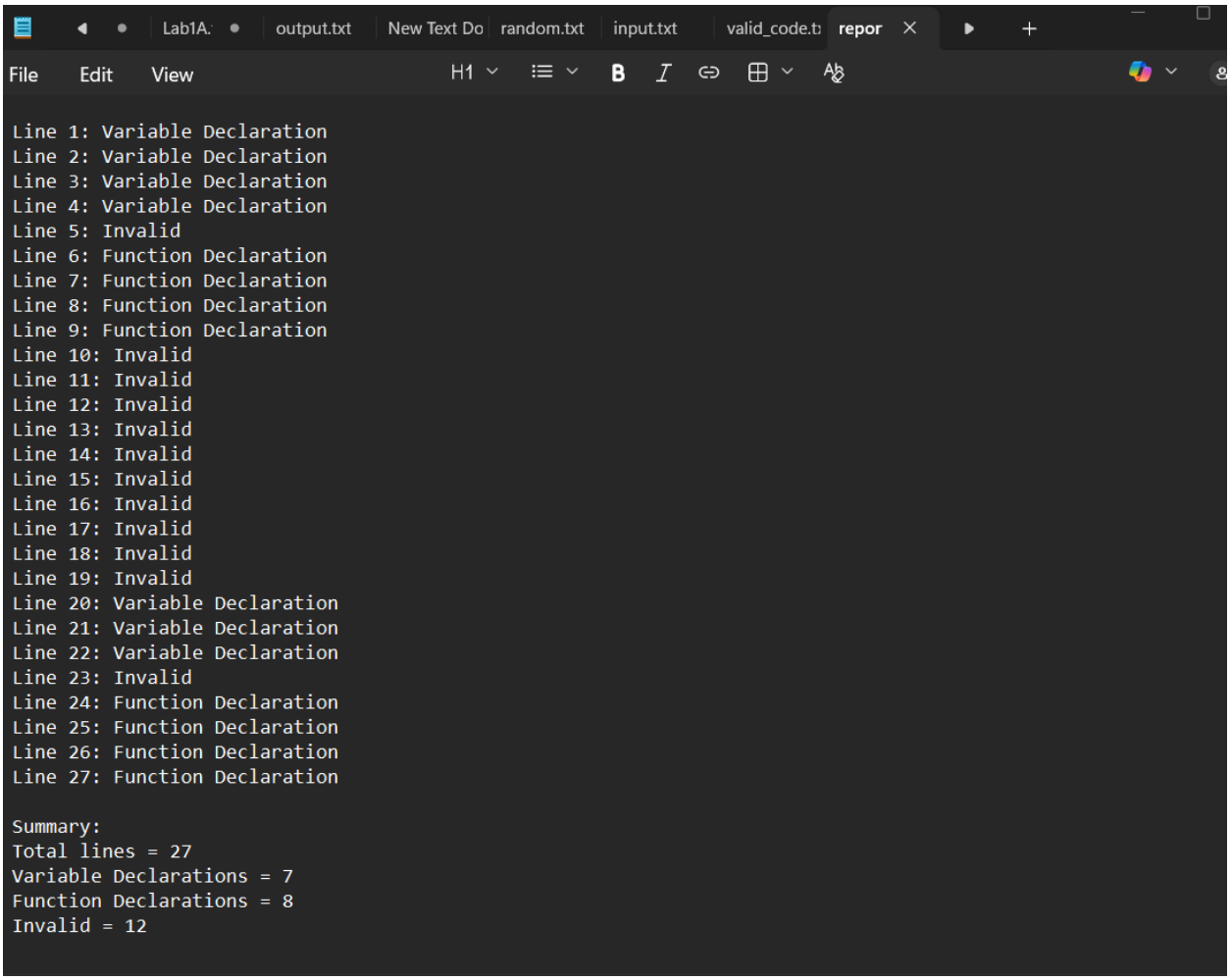
```
    while (getline(input, line))
    {
        lineNo++;
if (isVariable(line))
        {
            report << "Line " << lineNo << ": Variable Declaration\n";
            valid << line << endl;
            varCount++;
        }
    else if (isFunction(line))
        {
        report << "Line " << lineNo << ": Function Declaration\n";
        valid << line << endl;
        funCount++;
        }
    else
        {
        report << "Line " << lineNo << ": Invalid\n";
        invalid++;
        }
    }
    report << "\nSummary:\n";
    report << "Total lines = " << lineNo << endl;
    report << "Variable Declarations = " << varCount << endl;
    report << "Function Declarations = " << funCount << endl;
    report << "Invalid = " << invalid << endl;
    input.close();
    report.close();
```

```
    valid.close();
    cout << "report.txt and valid_code.txt successfully created!" << endl;
    return 0;
}
```

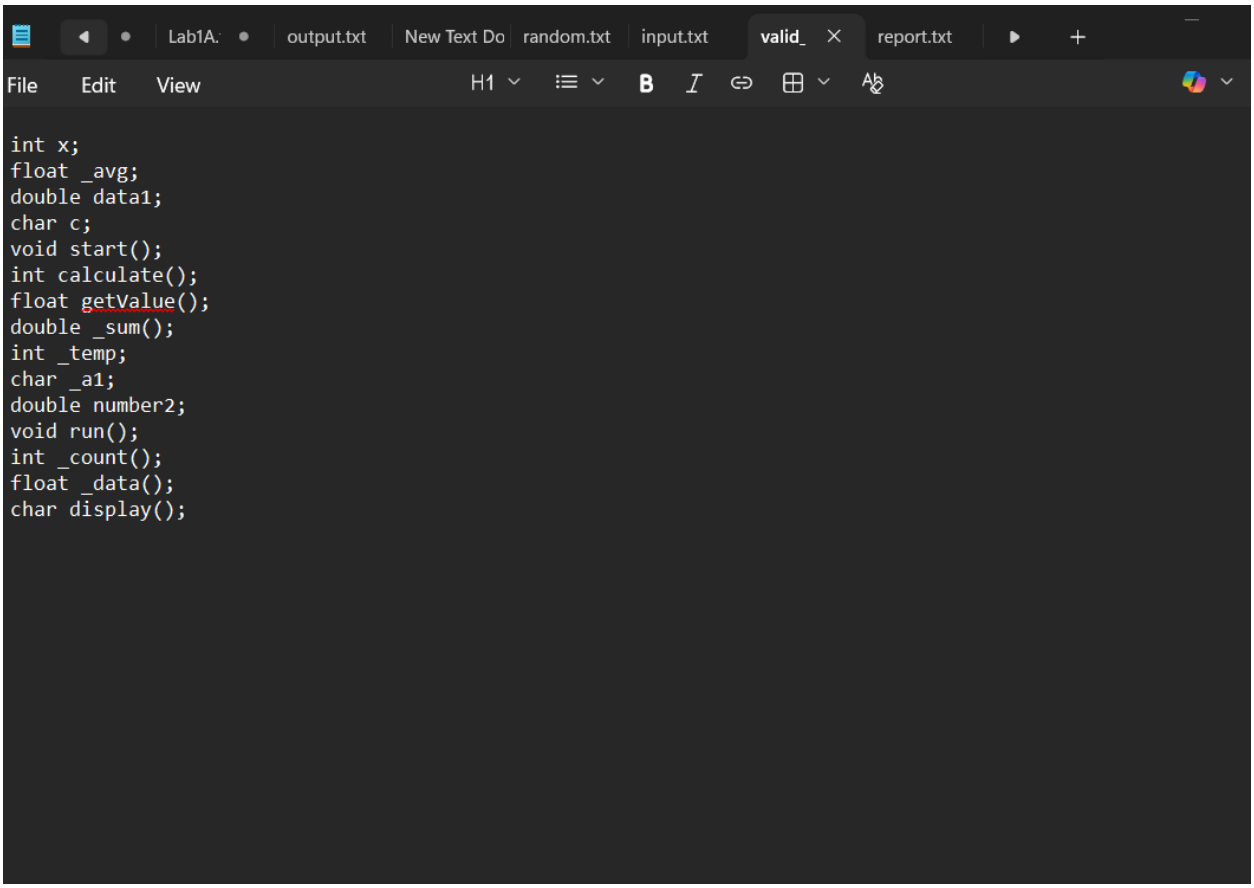
Input file ss:



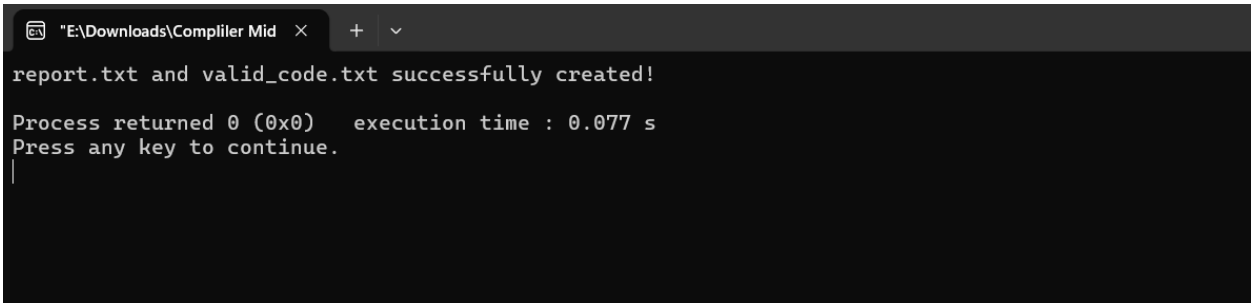
Report text file:



Valid_code text:



Cmd output:



////////////////////// **End**//////////////////////////////////